

Task 1 — Types of Operating Systems

The table compares three common operating systems based on their main features, target devices, and examples of applications commonly used on each.

Operating System	Main Features	Target Devices	Examples of Apps
Windows	User-friendly desktop interface, broad hardware/software compatibility, multitasking and extensive application support.	Desktops, laptops, workstations.	Microsoft Office, Chrome, VLC, Visual Studio.
Linux	Open-source, highly customizable, strong command-line tools, secure and widely used for servers and development.	Desktops, laptops, servers, embedded systems.	Firefox, LibreOffice, VS Code, GIMP.
Android	Mobile-focused operating system, touch interface, app ecosystem, notifications and device integration.	Smartphones, tablets, smart TVs and other mobile devices.	WhatsApp, Instagram, YouTube, Google Maps.

Summary: Windows is widely used for personal computers, Linux is popular for customization, development and servers, and Android is designed primarily for mobile devices.

Task 2 — Operating Systems on My Devices

The operating systems used on the devices considered for this assignment are:

Device	Operating System	Two Features Affecting Daily App Use
Laptop/Desktop	Windows 11 Pro	1. Wide software compatibility allows applications such as Chrome, Office and development tools to run easily. 2. Multitasking allows several applications and browser windows to run together.
Phone	Android	1. App permissions and notification controls affect how apps such as Instagram, WhatsApp and Zomato access device features and communicate with the user. 2. Android's touch-first interface and mobile app ecosystem make social, food-delivery and communication apps convenient to use.

Daily-use examples: Instagram and Zomato depend on internet connectivity, notifications, permissions and smooth multitasking or app switching.

Task 3 — Security Risks in Mobile Operating Systems

Mobile operating systems such as Android and iOS face several security risks that can directly affect apps used for communication, payments and online services.

Security Risk	Explanation	Possible Impact on Apps
Malicious or Fake Apps	Attackers may distribute apps that imitate legitimate applications or contain malicious code.	A fake WhatsApp, Paytm or banking-related app could attempt to steal login information, messages or financial data.
Excessive App Permissions	An app may request access to sensitive resources such as contacts, location, camera, microphone or files without a strong need.	Unnecessary access can expose personal information and reduce privacy when using apps such as WhatsApp or payment applications.
Phishing and Social Engineering	Users may receive fake links, messages or login pages designed to trick them into revealing credentials or payment information.	A user could enter a password, OTP or other sensitive information into a fraudulent page pretending to be a trusted service.

Basic protection: install apps from trusted sources, review permissions, keep the operating system and apps updated, and avoid suspicious links or login pages.

Task 4 — Cloud-Based Operating System vs Traditional Desktop OS

A cloud-based operating system or virtual desktop environment relies more heavily on remote services, cloud storage and internet connectivity than a traditional desktop operating system installed directly on a computer.

Point	Cloud-Based OS / Virtual Desktop	Traditional Desktop OS
Where work runs	Some or much of the computing environment and data may be provided through remote/cloud services.	Applications and most system processing run locally on the computer.
Internet dependency	Internet access is often important for accessing cloud services and remote resources.	Many local applications can continue working without an internet connection.
Updates & management	Cloud services can centralize management and updates for the environment.	The local device normally manages its own OS and application updates.
Example	ChromeOS and browser-based cloud applications; virtual desktop services.	Windows or Linux installed directly on a laptop/desktop.

Example of changed app use: On a cloud-focused system, a student may use a browser-based video service such as YouTube without installing a traditional desktop application, while files and work can also be accessed through cloud services. A traditional desktop can run locally installed applications and store files directly on its internal drive.